



ATERA ANALYTICS

**Optimising Technology
Feasibility for Autonomous
and Remotely Controlled
Vehicles**

Project Progression Q1

DATE

August 2025

Introduction

Introductions

Minutes of last meeting – at least review actions

IUK updates – if any

Technical progress updates, next steps and demos

- a. Verbal progress update & demos
- b. Q progress, talk through Progress report

Project Management update, (plan, deliverables and milestone register review)

- a. Review updates to L2 plan,
- b. Gantt Chart
- c. Deliverables register
- d. Milestone register

Financial review – review claim

Exploitation Planning – Exploitation plan updates

Risk register

Summary of new actions

Any other business

Monitoring Officer feedback

Confirm next mtg dates



Cloud

Cloud –
WPs 1,4,5,6

Google Cloud
Developer Environment

Project Management

AI and ML on routes
assessment

POSIT and Plotly Cloud
for Dashboards

Platform

Integrated Cloud Infrastructure: Google Cloud Platform provides scalable AI processing and data storage, Atlassian Trello manages collaborative workflows, ArcGIS delivers spatial intelligence for UK infrastructure mapping and route optimisation. Cloud platform evolves from basic hosting to sophisticated modelling.

Route Viability Visualisations: Real-time color-coded route scoring system displaying infrastructure readiness, safety assessments, and autonomous vehicle compatibility ratings with predictive journey intelligence and alternative route recommendations.

Planning Intelligence: Comprehensive assessment and monitoring of infrastructure performance analytics, safety incident coordination, commercial metrics tracking, and forecasting for operational optimisation of the roads to enable deployment of autonomous vehicles.

Databases

WPs 2,3,4,5,6

DAFNI Infrastructure for
Transport

Atera Analytics Data
on UK Road Infrastructure

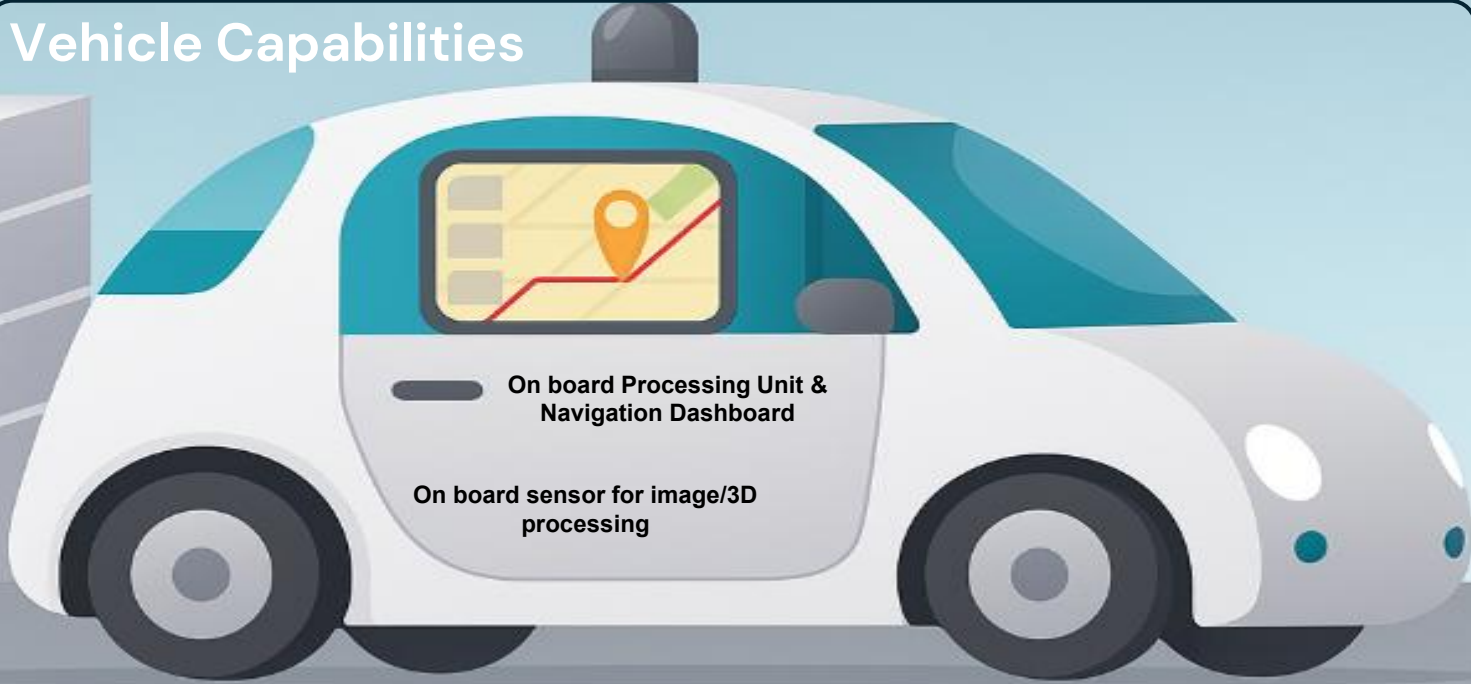
ArcGIS and
Purchased Data

Python Based
APIs Accessed Data

All data first hosted in
Atera Analytics Infrastructure

Some sources will be extracted
and hosted on the cloud.

Vehicle Capabilities

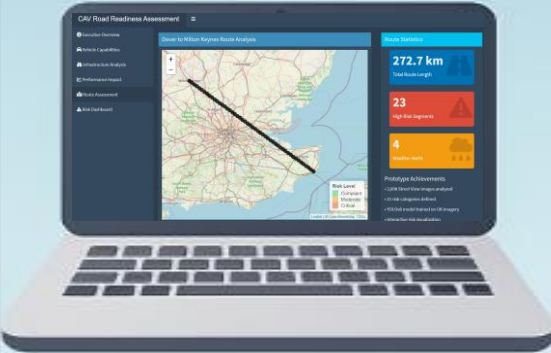


Onboard processing unit primarily required for WP6 (advanced CCAV integration) with some involvement in WP5 and WP7) Relevant insights could include:

- Road Quality Scoring: Dynamic 1-10 ratings for upcoming road segments based on surface conditions, maintenance status, and AV suitability
- Infrastructure Readiness Heat Map: Color-coded route visualization showing green (AV-ready), amber (caution required), and red (manual override recommended) sections

Reporting

Offline Visualisation and
Reporting –
WPs 2,3,4,6,7



Dashboards based on the overall
assessment of UK Infrastructure.

Visualisations - High importance for
stakeholder-facing work packages

Risks assessment by region

Work Package 1: Foundation

May 1, 2025 - Jul 31, 2025 | Status: COMPLETED

Lead: PM/Lead

Key Deliverables

- Project Management Tools (Jira, Trello)
- Code Repository Setup (GitHub)
- Development Environment Configuration
- Initial Cloud Infrastructure Setup
- Technical Hardware Procurement
- Software Licencing and Tool Selection
- Team Collaboration Framework

Progress Status

- ✓ All foundation tools operational
- ✓ Code repositories configured
- ✓ Development environments ready
- ✓ Team collaboration established

Task Breakdown

- 1.1 KOM - Zenzic and MSP
- 1.2 Software Tools Selection and Setup
- 1.3 Hardware Procurement and Configuration
- 1.4 AI Advisers Technical Meeting

Owner: PM/Lead

Dependencies: None

Focus: Technical Foundation Setup

Technical Foundation Established

- Development environments ready for Route 10 AI algorithms
- Code repositories configured for collaborative development
- Project management tools operational for team coordination

Technical Setup Approach

- Focus on development tools vs WP4 ongoing project coordination
- Establish technical foundation for AI development workflows
- Configure collaborative coding environment for multi-team project

CAM Infrastructure Assessment Platform - System Boundary

WP1: Foundation

Tools and Management
COMPLETED

WP2: Route 10

Under Review

WP3: Front End

In Progress

WP4: PM Setup

In Progress

WP5: Data Arch

In Progress

WP6: CAM

Not Started

WP7: Final

Not Started

Technical Foundation Achievements

Completed:

- Development tools and code repositories configured for collaborative AI development
- Technical infrastructure established enabling Route 10 algorithms and data processing

Enables Technical Work:

- Route 10 AI algorithm development with proper version control and collaboration
- Front-end dashboard development with integrated development workflows
- Data integration architecture with proper development and testing environments

Work Package 2: Route 10 Deliveries

Jul 1, 2025 - Oct 31, 2025 | Status: UNDER REVIEW

Lead: Route 10

Key Deliverables

- AI-based Infrastructure Assessment
- Large-scale Regional Analysis POC
- Local Route Assessment Tool
- Image Processing Algorithms
- Highway Safety Assessment
- Background Methodology Report
- Visual Output Demonstrations

Progress Status

- 📄 Deliverables under stakeholder review
- 📄 AI algorithms completed and validated
- 📄 Methodology report submitted
- 📄 Assessment tools operational

Task Breakdown

- 2.1 Large-scale Assessment POC
- 2.2 Local Assessment Algorithm
- 2.3 Methodology and Conclusions Report

Owner: Route 10 Collaborators

Dependencies: WP1 Foundation

Status: Under Stakeholder Review

Deliverables: AI Tools and Report

Completed Deliverables

- 📄 AI and image processing algorithms finalized
- 📄 Highway infrastructure assessment tool completed
- 📄 Comprehensive methodology report delivered

Technical Achievements

- 📄 Scalable AI modelling for UK roads infrastructure deployed
- 📄 Visual assessment of signs, dangers and safe areas completed
- 📄 Regional and local route analysis capabilities operational

CAM Infrastructure Assessment Platform - System Boundary

WP1: Foundation
COMPLETED

WP2: Route 10
AI Assessment Tools
UNDER REVIEW

WP3: Front End
In Progress

WP4: PM Setup
In Progress

WP5: Data Arch
In Progress

WP6: CAM
Not Started

WP7: Final
Not Started

Key Achievements and Integration

Completed:

- 📄 AI and image processing algorithms delivered for scalable UK infrastructure assessment
- 📄 Comprehensive methodology report with background analysis and conclusions finalized

Integration Ready:

- 📄 Visual assessment tools integrated with WP3 front-end for user interface display
- 📄 Route 10 outputs feeding into WP5 data architecture for comprehensive AV platform
- 📄 AI models ready for advanced integration in WP6 CAM testing phase

Work Package 3: Front End Application

May 1, 2025 - Oct 31, 2025 | Status: IN PROGRESS

Lead: PM/Lead

Key Deliverables

- Interactive Dashboard Interface
- Exploratory Data Analysis
- Visual UI Components
- POC Visualisation Prototypes
- Route 10 Report Integration
- AV Readiness Insights Display
- User Interface Design

Progress Status

- 🔥 POC visualisations completed
- 🔥 UI components development active
- 🔥 Data analysis framework ready
- 🔥 End-user engagement testing

Task Breakdown

- 3.1 Exploratory Data Analysis
- 3.2 User Interface Development
- 3.3 POC Visualisation Rendering
- 3.4 Route 10 Report Integration

Owner: PM/Lead and App Developer

Dependencies: WP1, WP2

Focus: Dashboard and Visualisation

Next Quarter Objectives

- Summarise road infrastructure data for AV patterns
- Deliver visual POC prototypes for stakeholder validation
- Complete interactive dashboard for end-user testing

Technical Approach

- Interactive visual interfaces for infrastructure insights
- Engaging UI components for road readiness visualisation
- Integration with Route 10 AI assessment outputs

CAM Infrastructure Assessment Platform - System Boundary

WP1: Foundation
COMPLETED

WP2: Route 10
Under Review

WP3: Front End
Dashboard and UI
IN PROGRESS

WP4: PM Setup
In Progress

WP5: Data Arch
In Progress

WP6: CAM
Not Started

WP7: Final
Not Started

Key Achievements and Next Steps

Completed:

- POC visualisations rendered and tailored for end-user engagement testing
- UI components built for visualising key insights on road infrastructure readiness

Next Quarter:

- Complete dashboard prototypes for stakeholder validation and iteration cycles
- Integrate Route 10 visual assessment outputs into interactive interface
- Deploy comprehensive data visualisation for AV infrastructure patterns

Work Package 4: Project Management and Setup

May 1, 2025 - Mar 31, 2026 | Status: IN PROGRESS

Lead: PM/Lead

Key Deliverables

- Scope and Goal Setting Framework
- Quarterly Review and Planning Sessions
- Stakeholder and Adviser Workshops
- Collaborative Web Tools Integration
- Training and Technical Events
- QRM Sessions and Progress Reporting
- Hardware Infrastructure Setup

Progress Status

- 🔥 Scope and roadmap defined
- 🔥 Operational readiness established
- 🔥 PM tools operational
- 🔥 Cloud infrastructure validated

Task Breakdown

- 4.1 Scope and Goal Setting
- 4.2 Quarterly Review and Planning
- 4.3 Adviser and Stakeholder Sessions
- 4.4 Web Tools Integration
- 4.5 Training and Technical Events
- 4.6 QRM 1 with Monitoring Officer
- 4.7 Hardware Setup

Owner: PM/Lead and Whole Team

Next Quarter Objectives

- Define scope, roadmap, and key milestones for AV evaluation
- Establish operational readiness via tooling and infrastructure
- Conduct quarterly planning sessions and stakeholder reviews

Management Approach

- Cross-project coordination and collaborative workflow management
- Regular stakeholder engagement and technical outreach events
- Quarterly review cycles ensuring alignment with project milestones

CAM Infrastructure Assessment Platform - System Boundary

WP1: Foundation
COMPLETED

WP2: Route 10
Under Review

WP3: Front End
In Progress

WP4: PM Setup
Management
IN PROGRESS

WP5: Data Arch
In Progress

WP6: CAM
Not Started

WP7: Final
Not Started

Key Achievements and Cross-Project Coordination

Completed:

- Project scope, roadmap, and key milestones defined for AV infrastructure evaluation
- Operational readiness established via tooling, training, and technical infrastructure

Ongoing:

- Quarterly planning sessions to adapt goals and review deliverables across all WPs
- Continuous stakeholder workshops and technical coordination throughout project lifecycle
- Hardware procurement and configuration for testing and development support

Work Package 5: Data Integration and Architecture

Jul 1, 2025 - Jan 31, 2026 | Status: IN PROGRESS
Lead: PM/Lead

Key Deliverables

- End-to-End Data Architecture
- CAM Operations Data Pipeline
- Digital Twin Model Definition
- External Data Source Integration
- AI Data Enrichment Platform
- Route Navigation Visualisation
- CAM Performance Data Validation

Progress Status

- 🔥 Data architecture designed
- 🔥 CAM data workflows enabled
- 🔥 POC visualisation prototyped
- 🔥 Basic functionality operational

Task Breakdown

- 5.1 CAM Operations Data Enablement
- 5.2 End-to-End Architecture Design
- 5.3 EV CAM Performance Validation
- 5.4 Digital Data Model Definition
- 5.5 External Data Integration
- 5.6 AI Data Enrichment
- 5.7 Route Navigation Visualisation
- Multi-team: PM/Lead, AI & Data Dev
- Dependencies: WP1, WP2

Next Quarter Objectives

- Integrate and validate external data sources
- Support simulation via enriched route data visualisation
- Deploy scalable data processing for AV operations

Technical Architecture

- Scalable data pipeline architecture for CAM operations
- AI & ML models for data enrichment and completeness
- Digital twin representing UK road infrastructure features

CAM Infrastructure Assessment Platform - System Boundary



Key Achievements & Next Steps

Completed:

- Robust AV data architecture designed and CAM data workflows enabled
- Basic functionality established for data processing pipeline operations

Next Quarter:

- Complete external data source integration to enrich AV readiness models
- Deploy AI data enrichment platform with ML models for enhanced accuracy
- Enable digital twin model and route navigation data visualisation capabilities

Work Package 6: CAM Integration and Testing

TBC - Mar 31, 2026 | Status: NOT STARTED
Lead: PM/Lead

Key Deliverables

- CAM Data API Connectivity
- 5G Data Processing Design
- ML Path Planning Models
- Remote Control Driving KPIs
- AI Model Validation Framework
- NVIDIA Digital Twin Module
- Automated API Data Access

Current Status

- II Not Started - Planned for Q2
- II Dependent on WP1-5 completion
- II Advanced AI capabilities phase
- II Integration and testing focus

Task Breakdown

- 6.1 Atera Analytics CAM Data API
- 6.2 CAM 5G Data Processing Design
- 6.3 AI Network Models Data Preparation
- 6.4 Machine Learning Path Planning
- 6.5 Remote Control Driving KPI Training
- 6.6 AI Models Validation
- 6.7 NVIDIA Digital Twin Deployment
- 6.8 Automated API Data Access

Multi-team: PM/Lead, AI, Dev, Contractor
Dependencies: All Previous WPs

Planned Next Quarter

- Ensure API connectivity for CAM data transmission
- Design navigation modules for on-device edge computing
- Initiate AI model training for autonomous vehicle KPIs

Advanced AI Integration

- Machine learning models for intelligent route planning and safety
- Real-world validation against CAM sensor and performance data
- NVIDIA Digital Twin deployment for advanced simulations

CAM Infrastructure Assessment Platform - System Boundary

WP1: Foundation
COMPLETED

WP2: Route 10
Under Review

WP3: Front End
In Progress

WP4: PM Setup
In Progress

WP5: Data Arch
In Progress

WP6: CAM
Integration and Testing
NOT STARTED

WP7: Final
Not Started

Dependencies and Readiness

Prerequisites:

- Data integration architecture from WP5 provides foundation for AI model training
- Route 10 AI algorithms from WP2 enable advanced path planning capabilities

Next Phase:

- Launch advanced AI integration with ML models for intelligent navigation
- Deploy NVIDIA Digital Twin module for comprehensive AV testing scenarios
- Validate trained models against real-world CAM sensor performance data

Work Package 7: Final Delivery

TBC - Mar 31, 2026 | Status: NOT STARTED

Lead: Whole Team

Key Deliverables

- Full Platform Implementation
- Integrated CAM Assessment System
- Comprehensive Project Report
- Final Stakeholder Presentations
- Commercial Platform Deployment
- Technical Documentation
- Knowledge Transfer Materials

Current Status

- || Not Started - Final Phase
- || Dependent on WP3-WP6 completion
- || Platform integration and deployment phase
- || Commercial readiness preparation

Task Breakdown

7.1 Full Platform Implementation

7.2 Project Reporting and Presentation

Owner: Whole Team

Dependencies: All Previous WPs

Status: Awaiting Prerequisites

Focus: Platform Deployment

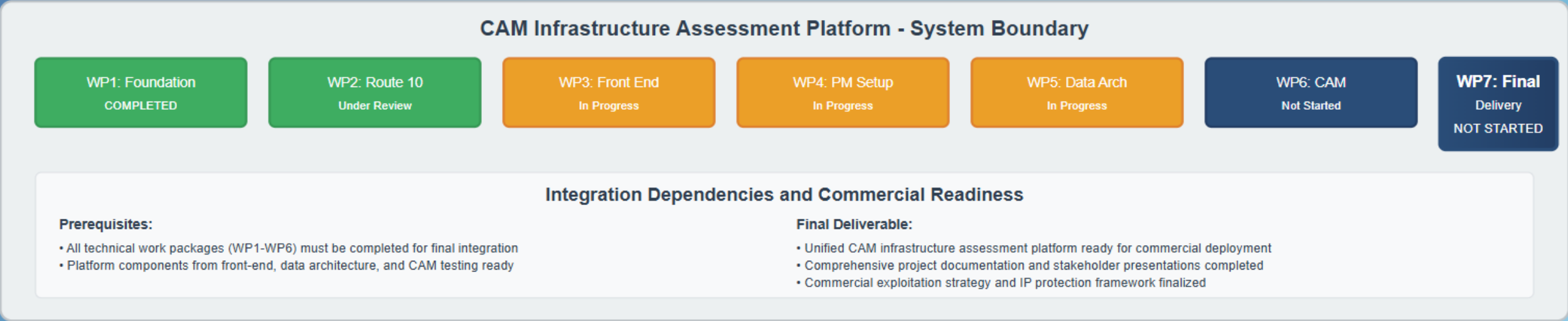
Goal: Commercial Readiness

Planned Final Phase

- Deploy comprehensive CAM infrastructure assessment platform
- Compile detailed project results and methodology documentation
- Deliver final stakeholder presentations and knowledge transfer

Commercial Platform Integration

- Unified platform integrating all WP outputs for market deployment
- Comprehensive documentation for commercial exploitation
- Final validation and testing of complete AV assessment system



Dashboard Prototype Demo – To be during the session

